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Dendrite short-circuit and fuse effect on Li/polymer/Li cells

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Abstract

We report on experimental and theoretical studies of dendritic growth in Li/polymer/Li symmetric cells. Potential evolution with time, impedance and in situ microscopy experiments enable to characterise the onset and evolution of dendrites. In particular we observe that dendrites may burn when a high enough current goes through them, a thermo-fusible effect predicted in a previous paper and confirmed by SEM experiments. We present a calculation that gives a quantitative description of this effect: our results enable to understand a series of experimental data published in the literature concerning impedance variations observed while cycling lithium-polymer cells.
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Keywords: Lithium-polymer battery; Lithium dendrite; Thermo-fusible effect; Impedance spectroscopy; In situ microscopy

1. Introduction: Li batteries and effect of dendrites

Despite recent progress, dendrite formation while charging lithium batteries with lithium metal negative electrode remains one of the major issues of this type of technology [1]. Even in the case of polymer or gel electrolytes, this phenomenon exists, although it was shown to be less important than in liquid electrolytes [2,3]. Dendrites are very detrimental to the operation of lithium batteries, for safety reasons, and because they limit the lifetime of batteries. Therefore, it is of great importance to understand how they appear, and their evolution.

We present in this paper a set of experimental results giving a coherent picture of dendritic growth in Li/polymer cells. Our analysis is based upon results obtained from four different techniques performed simultaneously, or in comparable situations: evolution of the cell potential with time, impedance measurements obtained while polarising the cells, in situ optical microscopy [4], SEM experiments [5]. In conjunction with a simple equivalent circuit calculation, these experiments enable to quantitatively describe the evolution of dendrites with time and a thermo-fusible effect predicted in Ref. [6], and confirmed by SEM experiments [5]. In particular, our analysis shows that dendritic growth induces a larger change on the impedance component due to the electrode/electrolyte interface than on the bulk impedance due to the electrolyte. This surprising result enables to explain other results published in the literature.

2. Experimental

The polymer electrolyte used in the experiments described in this paper consisted of polyethylene oxide (Mw = 300,000). The lithium salt was $LiN(CF_3SO_2)_2$ (abbreviated in LiTFSI), discovered by Armand et al. [7]. The salt concentration in terms of O/Li ratio was about 20. The polymer and the salt were mixed with acetone and acetonitrile: the mixture was heated until it became homogeneous. After having removed bubbles under vacuum, the electrolyte was spread on a spin coating machine and dried for at least 12 h under vacuum at 80 °C to remove the solvents. The obtained films were 35–80 μm thick.

Two different types of cells were used in this study. "Sandwich" cells were used for systematic electrochemical experiments and for SEM experiments [5]. Visualisation cells were used to follow in situ the formation and evolution of dendrites.

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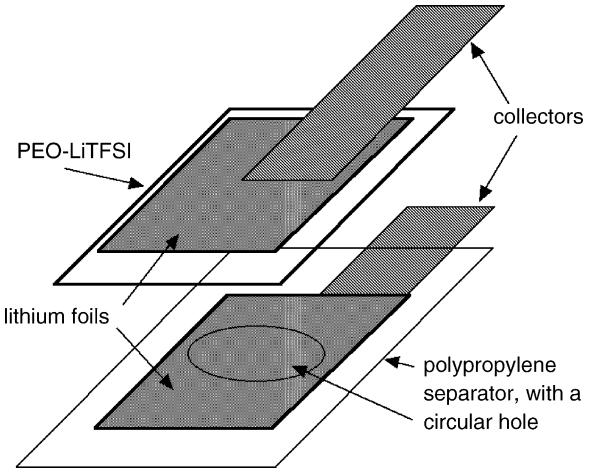


Fig. 1. Sandwich cell: the circular hole in the polypropylene separator defines the active part of the cell.

The sandwich cells essentially consisted of two lithium foils sandwiching a polymer electrolyte layer (Fig. 1). The inter-electrode distance was approximately determined by the film thickness, i.e. was in the range 35–100 μm.

The experimental set-up used for optical observation was detailed in Ref. [4]. A symmetric Li/electrolyte/Li configuration was prepared in a sealed glass cell and heated at 90 °C in a metallic box. The inter-electrode area was illuminated by infrared diodes; the pictures were taken by a black and white Charge Coupled Device (CCD) camera during polarisation, digitised with a Perceptics Pixelpipe framegrabber, and recorded on an Apple microcomputer. They were subjected to standard image analysis using the public domain NIH Image program (developed at the U.S. National Institutes of Health and available on the Internet at http://rsb.info.nih.gov/nih-image/).

Note that the geometry of the sandwich cells is markedly different from that of visualisation cells. The distance between the electrodes is 35–100 μm in sandwich cells whereas it is approximately 1 mm in visualisation cells. The section of the cells is 2–7 cm^2 in sandwich cells whereas it is approximately 0.05 cm^2 in visualisation cells.

Polarisations were performed in galvanostatic conditions using a Solartron 1286 electrochemical interface [8]. Impedance spectroscopy (IS) was performed with a Solartron 1255 frequency analyser, before and during polarisation, allowing to follow the evolution of both the electrolyte and the interface properties [9]. The impedance measurements performed during polarisations were done under the imposed constant current.

3. Experimental results

When polarising a cell with a binary electrolyte at high enough current density, the ionic concentrations in the vicinity of the negative electrode are expected to go to zero at the Sand time τ_S [10]. The Sand time varies as:

$$\tau_S = \pi D \left(\frac{z_c e C_{c0}}{2J} \right)^2 \left(\frac{\mu_a + \mu_c}{\mu_a} \right)^2 \quad (1)$$

where e is the elementary charge, μ_a and μ_c are the anionic and the cationic mobilities, respectively, D the ambipolar diffusion constant $D = (\mu_a D_c + \mu_c D_a)/(\mu_c + \mu_a)$, D_a and D_c being the anionic and cationic diffusion constants, respectively, J the current density, C_{c0} the initial cationic concentration and z_c is the cationic charge number. In the case where the anionic and cationic charge numbers z_a and z_c are equal $(\mu_a + \mu_c)/\mu_a = 1/t_a$, where t_a is the anionic transport number $(t_a = z_a\mu_a/(z_a\mu_a + z_c\mu_c))$.

In cells with a small inter-electrode distance, a different regime may be observed at low current density [4,11,12], below the limiting current density; after a time t_{eq} depending on the diffusion constant D and on the inter-electrode distance l, the system tends to a steady state, where the concentration varies almost linearly from $C_- = C_0 - \Delta C_-$ at the negative electrode to $C_+ = C_0 + \Delta C_+$ at the positive electrode.

The concentration gradient is given by [11]:

$$\frac{dC}{dx} = \frac{J\mu_a}{De(\mu_a + \mu_c)} \quad (2)$$

From Eq. (2), one can easily show [4] that

$$-\Delta C_- \sim \Delta C_+ \sim \frac{\mu_a}{\mu_a + \mu_c} \frac{Jl}{eD} \quad (3)$$

The time t_{eq} is very close to the diffusion time t_D on the distance l, $t_D \sim l^2/D$.

The crossover between the two regimes is at the current density $J^* = 2eC_0D(\mu_a + \mu_c)/\mu_a l$ [4,11,12]. This roughly corresponds to the current density J where the Sand time is equal to the diffusion time on the distance l, i.e. $\tau_S \sim t_D$.

All experiments reported in this paper were carried out in the low current density regime.

3.1. Variation of the voltage—sandwich cells

A typical variation of the voltage measured across a cell subjected to a constant current is shown in Fig. 2. When one applies a constant current through a Li/POE/Li cell, one usually observes the following behaviour [4,13]:

- The voltage rapidly increases to a value $V_0 \sim (R_e + R_i)I$, where R_e and R_i are the bulk and interface resistance, respectively [9]. Then the voltage increases more slowly to a stationary value. This latter behaviour is related to the establishment of the concentration gradient through the cell. In principle, this steady state remains until the whole lithium has been removed from the anode; then the potential eventually diverges, due to conductivity failure of the electrode. This happens after a time t_{ex}.

$$t_{ex} = \frac{deNT_{Li}}{M_{Li}J} \quad (4)$$

  where d is the lithium density, N the Avogadro number, T_{Li} the thickness of the lithium electrode and M_{Li} is the lithium molar mass [14].
- However, another behaviour is observed in almost all the cells we have tested. After a time t_{cc}, a rapid decrease in the potential

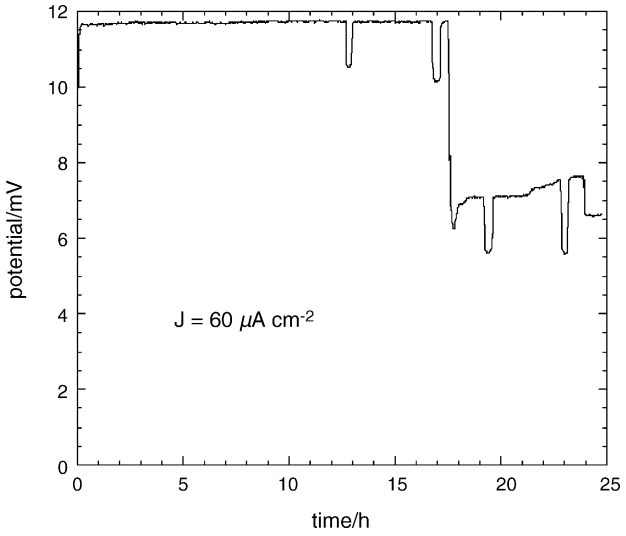


Fig. 2. The "fuse effect": time variation of the potential. The diameter of the active part of the cell is 3 cm. The inter-electrode distance is 80 μm.

is observed (Fig. 2) that we ascribed to short-circuiting of the cell by one or several dendrites. Although a relatively large dispersion of times t_{cc} was observed, we found that the time t_{cc} roughly varied with the applied current density according to:

$$t_{cc} \sim J^{-2} \quad (5)$$

More precisely t_{cc} had a behaviour very similar to the Sand time τ_S (Eq. (1)) [15].

In fact, this behaviour was very surprising, because, at low current density, no Sand's behaviour is expected [15]. To explain this phenomenon, we proposed a model [15,16] taking into account the non-uniformity of the lithium–electrolyte interface and the small inter-electrode distance. According to this model, due to variations in the local current density at the lithium/electrolyte interfaces, concentration gradients developed in the cell, leading to a "transverse Sand/Chazalviel" behaviour. If we divide the inter-electrode region in small boxes of size l, the current density may very probably vary from one box to the other. The current entering in a small box unit may even be different from the current exiting the box on the opposite interface. Local sources and sinks for current density are thereby created. Now, transversal diffusion tends to limit these local concentration fluctuations. Then, the situation is almost the same as in the usual Sand's model, but for a concentration evolution in a direction parallel to the electrodes. According to this model, t_{cc} is not exactly equal to τ_S, because the variation of J, rather than J determines its value.

After this short-circuit effect, usually a very perturbed behaviour is observed, with large random fluctuations of the potential. However, before this happens, we very often observe a return to a normal behaviour (Fig. 2) [5]. Eventually a second decrease in the potential is observed after a time that can be relatively long. Actually, both the duration of the first short-circuit and of the second "normal behaviour" phase may vary over a rather wide range (however, the duration of the second

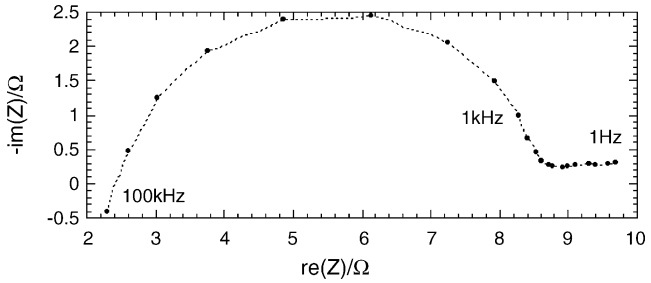


Fig. 3. Typical impedance diagram of a sandwich cell. The diameter of the active part of the cell is 3 cm. The inter-electrode distance is 80 μm.

phase is usually much shorter than t_{cc}). Some times the same phenomenon may be repeated a few times, but usually the very disturbed behaviour appears very soon.

A different explanation has been given for the variation of potential with time [17,18]. The authors observed that this decrease in the potential was essentially due to a decrease in the lithium/electrolyte interface contribution to the impedance. Then they assumed that the decrease in this impedance was due to local breakdowns of the passivation layer.

We will try here to analyse our experimental results and investigate how they may be interpreted with regard to these different models.

3.2. Effect on R_e, R_i

We have measured the variations of the cell impedance as well as the time evolution of the potential of a cell polarised at a constant current I. A typical impedance diagram is shown in Fig. 3. A detailed description of the impedance diagrams of Li/polymer/Li cells was given in Ref. [9]. The variation of the impedance with time is shown in Fig. 4. In this figure we report the values $I \times R_e$ and $I \times R_i$, as well as the value of the cell potential, for comparison. As mentioned by other authors [17,18], the variations of the overall impedance are mainly due to fluctuations of R_i. This is confirmed by the diagrams shown in Fig. 5,

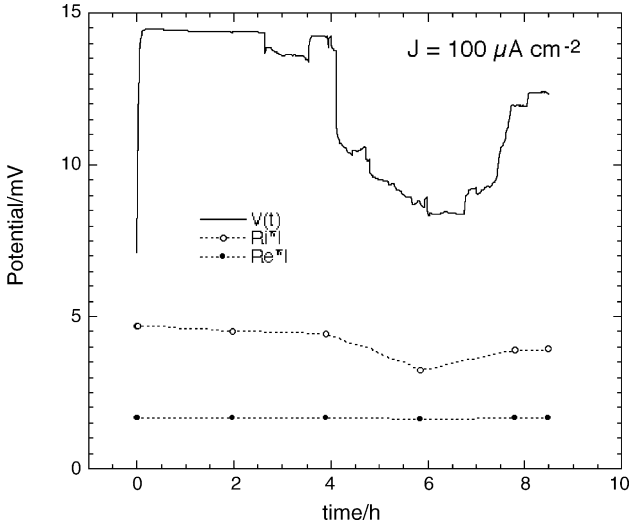


Fig. 4. Variation of the potential, of $I \times R_e$ and $I \times R_i$ with time. Same cell as in Fig. 3.

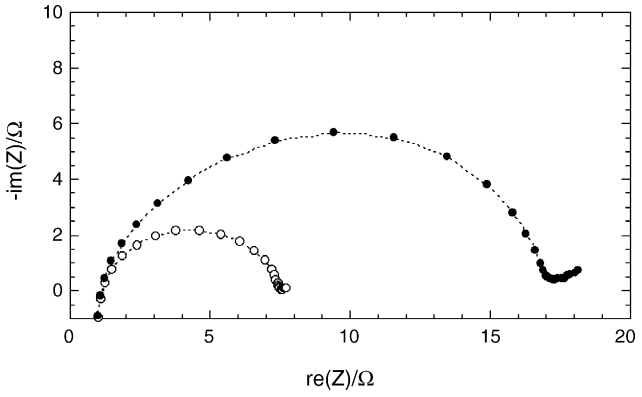


Fig. 5. Impedance diagrams recorded before (closed circles) and at the end (open circles) of a polarisation. The diameter of the active part of the cell is 1.6 cm. The inter-electrode distance is 35 μm.

which were recorded for a cell polarised at $J = 175\ \mu A\ cm^{-2}$ just before and at the end of the polarisation. Again we observe that R_e remains almost constant, whereas R_i decreases significantly.

3.3. Precise shape of V(t) (Fig. 6)

The evolution of $V(t)$ has a very similar behaviour at steps 1, 2 and 3 (see Fig. 6). The potential increases (1 and 3), respectively, decreases (2) to a value V_1, V_2 or V_3, then tends to a stationary value in a time equal to the diffusion time $t_D = l^2/D$, where l is the inter-electrode distance and D the ambipolar diffusion coefficient. We interpret this behaviour in terms of a two-step evolution: the rapid increase (resp. decrease) corresponds to an Ohmic part: $\Delta V = R\,\Delta I$. The subsequent evolution of the potential to a stationary value may be ascribed to the formation (1) or change of the concentration gradient in the electrolyte due to the effective current (the part of the current not flowing through the dendrite) (2 and 3).

3.4. Visualisation experiments

The onset and the evolution of a dendrite were followed using conventional optical microscopy [4,14] and SEM [5]. The variation of potential with time is shown in Fig. 7a, whereas three different steps of the growth are shown in Fig. 7b. Note that the geometry of the visualisation cell is very different from the geometry of sandwich cells. This explains the different values obtained for the cell potential, in comparison with those reported in Figs. 2, 4 and 6. It also explains the difference between the values obtained for equilibration time t_{eq}: this is expected to be much longer in visualisation cells because the distance between the electrodes is much larger. Therefore, t_{eq} is expected to vary as the square of the inter-electrode distance.

We observe in Fig. 7a (arrow 1) that the onset of dendritic growth is hardly visible on the potential evolution. This effect is expected to be even more pronounced in the case of sandwich cell, where the ratio between the resistance of a single dendrite and the bulk resistance of the cell should be much larger. This will be presented in more details below. The evolution of potential with time is only slightly affected by the growth of the

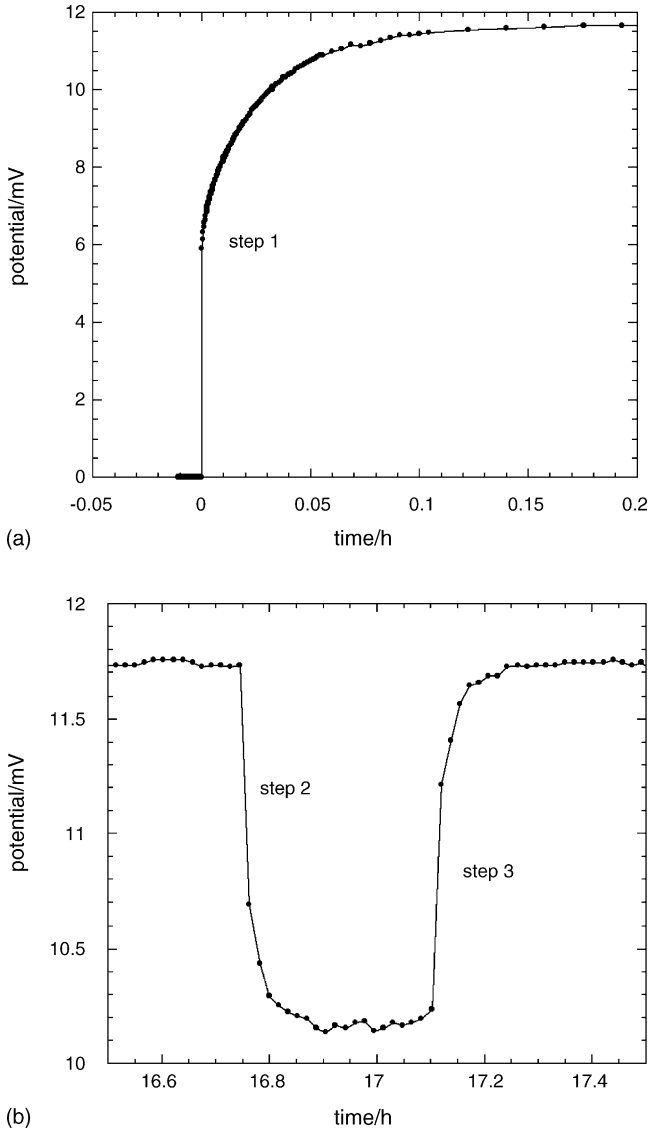


Fig. 6. Enlarged views of the time variation of the potential shown in Fig. 2: (a) just after the onset of polarisation (step 1) and (b) at the beginning (step 2) and at the end (step 3) of the short-circuit.

dendrite; in fact, the short-circuit is only clearly visible when the dendrite has reached the opposite electrode. Then a marked decrease in the potential is observed (arrow 2 in Fig. 7a), followed by an erratic evolution and a clear short-circuit.

Dollé et al. performed SEM observations [5] on Li symmetric cells just after a short-circuit was detected (Fig. 8). They noted that, whereas nothing was observed before the polarisation, a dendrite that went through the polymer was observed just after the short-circuit. Moreover, it was also noted that the polymer was different at the dendrite extremity, as if it had burnt when the dendrite short-circuited the cell. From this observation the hypothesis of a thermo-fusible effect was confirmed.

4. A simple equivalent circuit calculation

We present now a calculation of an equivalent circuit, which enables to understand the observed behaviours. Schematically

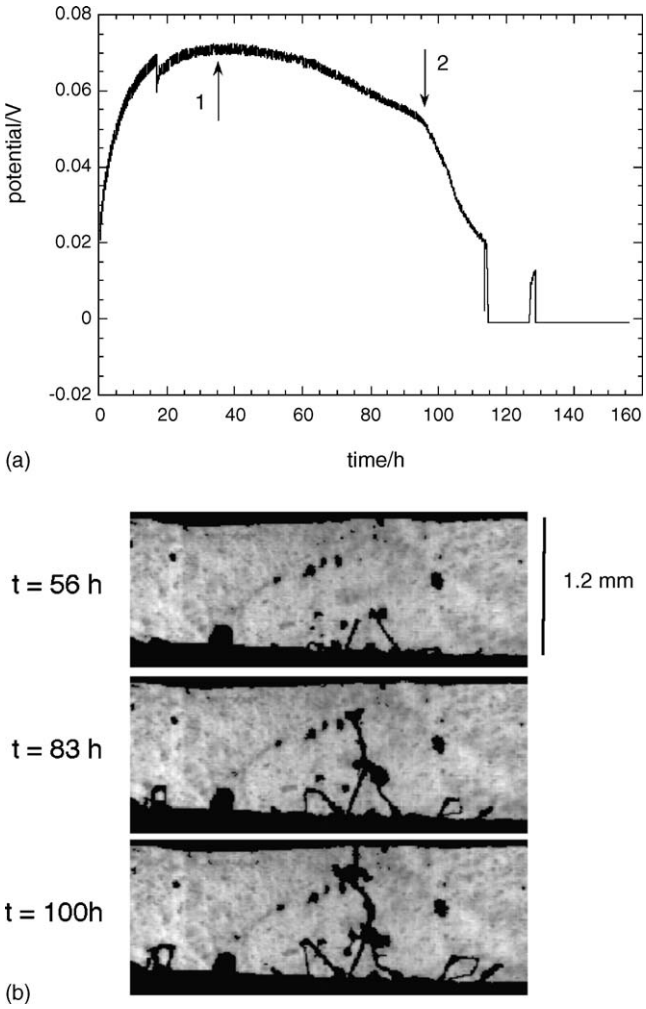


Fig. 7. Dendritic growth in a visualisation cell. (a) Variation of the cell potential with time. (b) Three steps of the growth (from Ref. [4]). The inter-electrode distance is 1.2 mm.

we represent our system as a circuit (Fig. 9) consisting of two components in parallel: one represents the original electrolyte consisting of a resistance R_1 in series with two RC circuits R_2C_2. The RC circuits represent the two electrode/electrolyte interfaces (passivation layers), the resistance R_1 holds for the electrolyte. Here we neglect a capacitance in parallel with the resistance R_1:

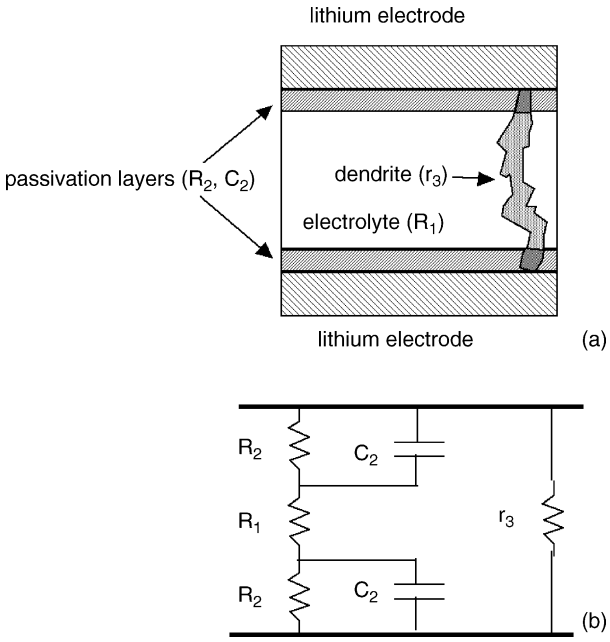


Fig. 9. Description of the Li/PEO/Li cell in terms of a simple circuit diagram.

this capacitance is expected to be very small, hence, its influence is outside the frequency range investigated in our experiments. In the case of our "sandwich cells" with the PEO electrolyte, the two resistances R_1 and R_2 are of the same order of magnitude. We consider the case where a dendrite has short-circuited the cell: we represent this dendrite by a pure r_3 resistance. The resistance r_3 may vary over a very wide range, but typically, it is of the same order of magnitude as R_1 or R_2. Starting from these hypothesis, we can now calculate the overall impedance of the above circuit.

Initially, i.e. before the dendrite has grown, the impedance Z of the cell is simply:

$$Z = R_1 + 2R_2 \frac{1 - jC_2R_2\omega}{1 + C_2^2R_2^2\omega^2} \tag{6}$$

With the resistance r_3 in parallel, this becomes:

$$Z' = \frac{R_1r_3(1 + C_2^2R_2^2\omega^2) + 2R_2r_3 - 2jr_3R_2^2C_2\omega}{(R_1 + r_3)(1 + C_2^2R_2^2\omega^2) + 2R_2(1 - jC_2R_2\omega)} \tag{7}$$

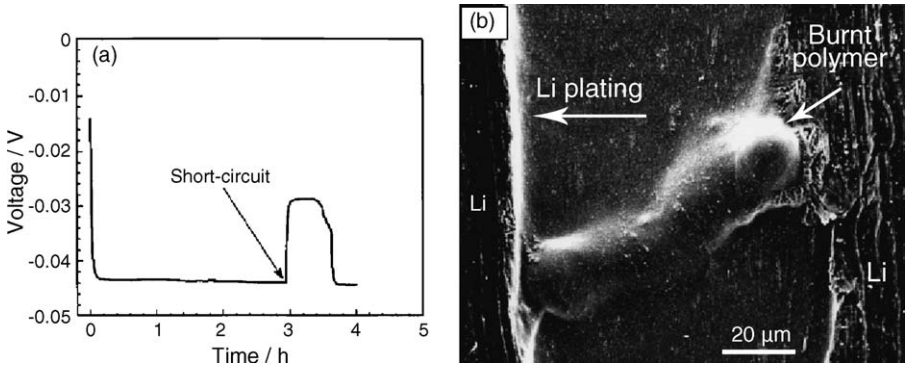


Fig. 8. SEM experiment (from Ref. [5]): (a) variation of the potential with time and (b) SEM picture of a dendrite, just after it has crossed the electrolyte polymer.

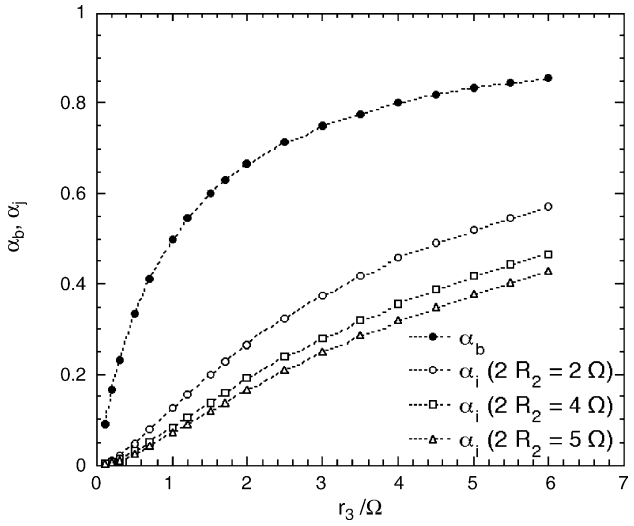


Fig. 10. Variation of the bulk ratio α_b and of the interface ratio α_i with r_3, for three different values of the initial value of the interface resistance $2R_2$.

its theoretical resistance: about 30 mΩ. Compared to usual cell resistances, this should give a clear short-circuit that we do not see in Fig. 8a: V is only reduced by approximately 40%. This confirms that the dendrite does not completely short-circuit the cell, probably due to the bad quality of the electrical contact between the dendrite and the electrodes. This last point is confirmed by the fact that burned polymer is seen at one of the extremities of the dendrite.

5. Conclusion

We report in this paper a detailed study of the formation and evolution of dendrites in a symmetric lithium/polymer electrolyte/lithium cell. We compare several types of experiments: electrochemical measurements in a "sandwich" cell, and in situ visualisation experiments using conventional optical microscopy. The results were analysed in terms of a simple model for the impedance of the cell, and confirmed by SEM data obtained by Dollé et al. [5].

$$Z' = \frac{[r_3R_1(1+\omega^2C_2^2R_2^2)+2r_3R_2][(R_1+r_3)(1+\omega^2C_2^2R_2^2)+2R_2]+4\omega^2C_2^2R_2^4r_3-2j\omega C_2R_2^2r_3^2(1+\omega^2C_2^2R_2^2)}{[(R_1+r_3)(1+\omega^2C_2^2R_2^2)+2R_2]^2+4\omega^2C_2^2R_2^4} \quad (8)$$

At low and high frequencies, this becomes:

$$\omega \to 0: \quad Z'_0 = \frac{r_3(R_1+2R_2)}{R_1+2R_2+r_3} = \left(\frac{1}{R_1+2R_2}+\frac{1}{r_3}\right)^{-1} \quad (9)$$

$$\omega \to \infty: \quad Z'_\infty = \frac{r_3R_1}{R_1+r_3} = \left(\frac{1}{R_1}+\frac{1}{r_3}\right)^{-1} \quad (10)$$

Now from these two limits, one can approximately [9] extract the bulk and the interface resistance of the cell: $R_b = Z'_\infty$, and $R_i = Z'_0 - Z'_\infty$.

Hence, $R_b = r_3R_1/(r_3+R_1)$ and $R_i = r_3(R_1+2R_2)/(r_3+R_1+2R_2) - r_3R_1/(r_3+R_1)$.

Let us compare these values with the initial values, before the dendrite has appeared: we call α_i (resp. α_b) the ratio between the R_i (resp. the R_b) with and without the dendrite. One has:

$$\alpha_i = \frac{r_3^2}{(r_3+R_1)(r_3+2R_2+R_1)} \quad (11)$$

$$\alpha_b = \frac{r_3}{r_3+R_1} \quad (12)$$

Fig. 10 shows the variation of these ratios with r_3 for different values of $2R_2$. It appears that, due to the different dependence on r_3, α_i is much more affected by the dendrite than α_b. We believe that the apparent variation of the interface resistance, which has been observed by several authors [17,18], is due to the short-circuiting of the cell by a dendrite.

In the example shown in Fig. 4, we are now able to calculate the value of the resistance of the dendrite: we find $r_3 \sim 30\,\Omega$. If we suppose the dendrite is 100 μm long, this would be compatible with a dendrite diameter of approximately 0.6 μm. By comparison, the dendrite shown in Fig. 8b [5] has a diameter of 20 μm, which is compatible with other observations [7]. From the diameter of the dendrite seen in Fig. 8 we can also deduce

Our main results may be summarised as follows:

- The influence of the dendrite on the overall impedance is negligible, but when the dendrite has completely short-circuited the cell. Very probably, several dendrites may grow simultaneously at a very close velocity. However, also very probably, only one is first reaching the opposite electrode: we then believe that this first dendrite will act as a fuse, before other dendrites can connect the opposite electrodes (which will eventually occur as evidenced by the erratic potential evolution which is observed later).
- The apparent variation of the interface impedance is larger than that of the bulk impedance.
- From our results we are able to determine the value of a dendrite resistance in a given situation. This gives an order of magnitude estimate of the dendrite resistance observed in Li/polymer electrolyte systems.

In particular, our results enable to give a clear description of dendrites growing in Li/polymer electrolyte/Li cells. They explain several data published in the literature and that were previously misunderstood [17,18].

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